

Datavis_report

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Summary

Colors play a significant role in how we perceive life and the world around us. For many of us, blue represents peace, while red brings to mind love or danger. In psychology, the interaction of colors on emotional responses—a topic debated for over a century and not yet fully resolved—remains a subject of curiosity. The relationship between colors and emotions has been studied for more than a hundred years. Taking the example of ‘Red’, numerous metaphors related to color, such as ‘seeing red’ (i.e., being angry), exist in various languages, including English, German, and Chinese. Complementary to linguistic metaphors, it has been reported that red is associated with danger or anger in the USA, the United Kingdom, France, and China. In addition to negative connotations, red is also seen to be associated with positive emotions in Spain, the United Kingdom, and China. As we can see from these examples, a single color can carry various associations that may or may not be the same for different countries.

This thought is what pushed and motivated me toward this project topic. Do colors make us feel the same way in our culture as they do in other cultures? Are the feelings evoked by colors universal perceptions stemming from biological human nature, or are they perceptions and feelings formed as a result of the influence of the culture of the society we were born and raised in? This question was the biggest factor in my choosing this topic. Therefore, in my project, I chose Eastern and Western cultures—two opposite poles—to capture the sharpest difference in color perception and to distinguish the differences. This is because, in science, the greatest differences emerge at opposite ends. I took care to select data containing countries with these contrasting cultures to capture the sharpest difference in color perception. Thus, I aim to see whether a person’s perception of colors is influenced by individual preferences, universal factors, or cultural differences.

The dataset is a highly reliable one, conducted by Jonauskaitė et al. in 2019, peer-reviewed internationally, and published in the journal Royal Society Open Science. The data presented here were extracted from an ongoing online survey. It covers 8,532 observations collected from four main cultural regions: China (Eastern culture), Germany, the United Kingdom (Western culture), and Greece. Online, 711 participants from China, Germany, Greece, and the United Kingdom associated 12 color terms in their native languages with 20 separate emotion terms such as anger, joy, hate, regret, pride, shame, disgust, contempt, disappointment, and compassion. Each emotion variable is expressed with a numerical score ranging from 0 (no relationship) to 5 (strongest relationship).

Dataset Description

The dataset used in this project is the supplementary material of the article titled “A machine learning approach to quantify the specificity of colour–emotion associations and their cultural differences,” conducted by Jonauskaitė et al. (2019) and published in the internationally internationally respected journal Royal Society Open Science. The dataset possesses high scientific reliability as it has undergone peer review.

The analyzed file consists of 8,532 observations (rows) and 23 variables. The data includes responses from 711 participants representing four different cultural regions. These countries and their codes are: China (cn), Germany (de), United Kingdom (gb), and Greece (gr).

Variables: Within the scope of the analysis, 12 basic color terms were presented to the participants: Black, Blue, Brown, Green, Grey, Orange, Pink, Purple, Red, Turquoise, White, and Yellow.

20 emotion variables containing intensity scores between 0 (no relationship) and 5 (strongest relationship) given by participants to these colors were used:

Positive: Admiration, Amusement, Compassion, Contentment, Interest, Joy, Love, Pleasure, Pride, Relief.

Negative: Anger, Contempt, Disappointment, Disgust, Fear, Guilt, Hate, Regret, Sadness, Shame.

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